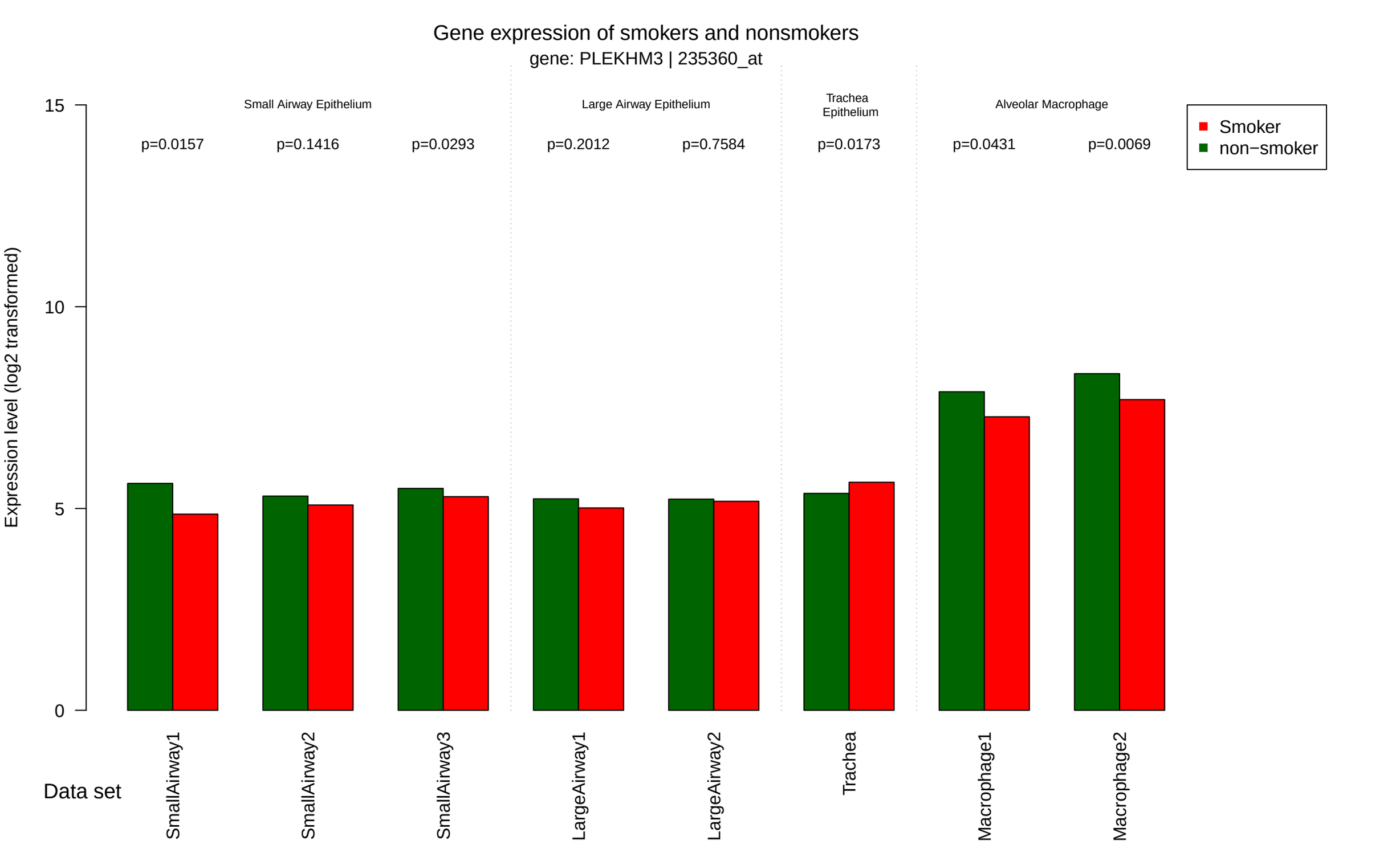
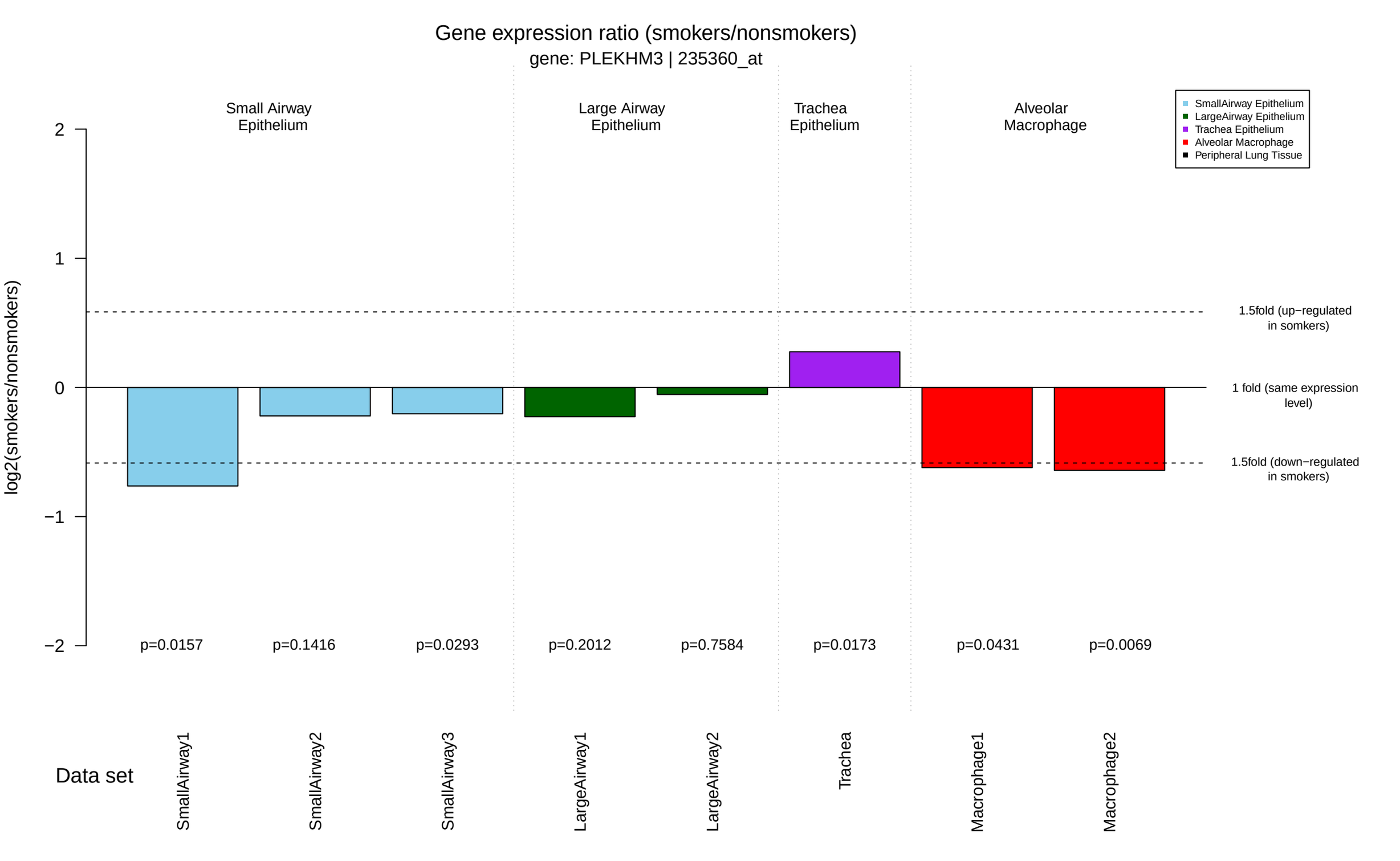
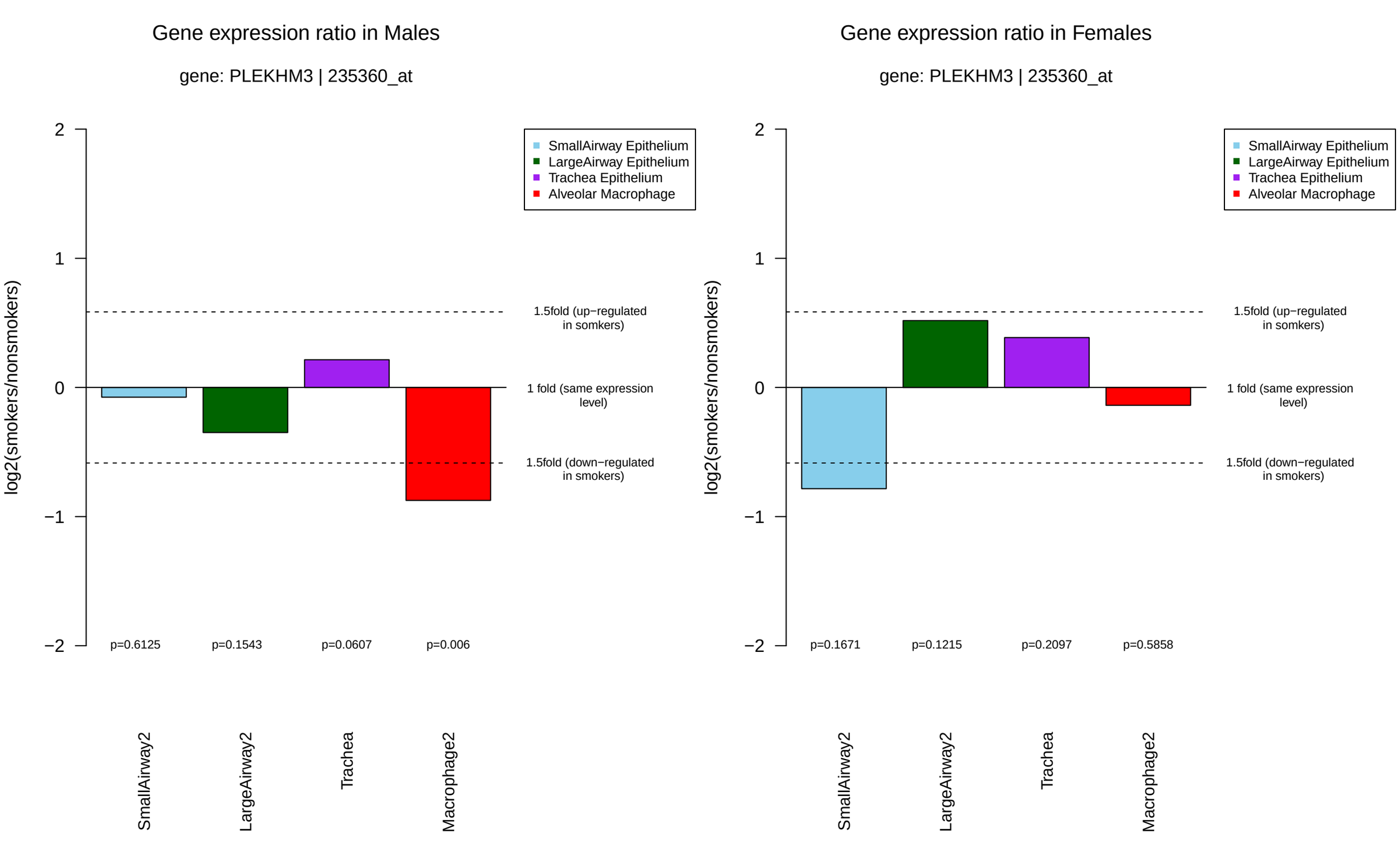
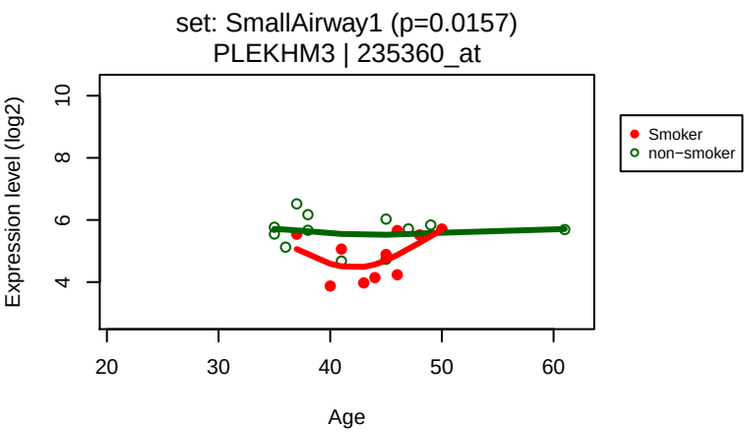
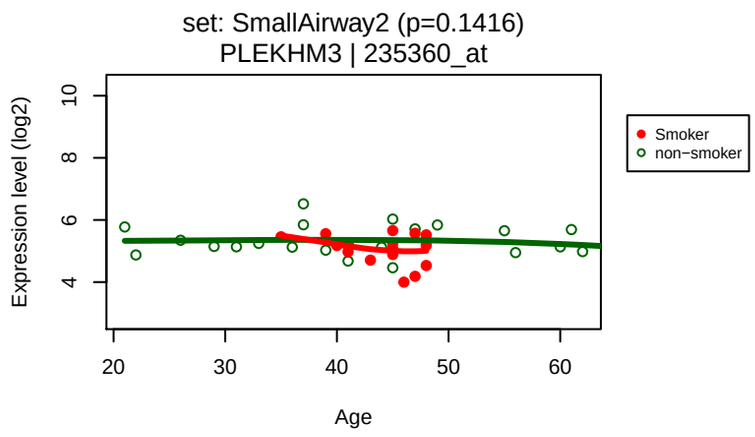
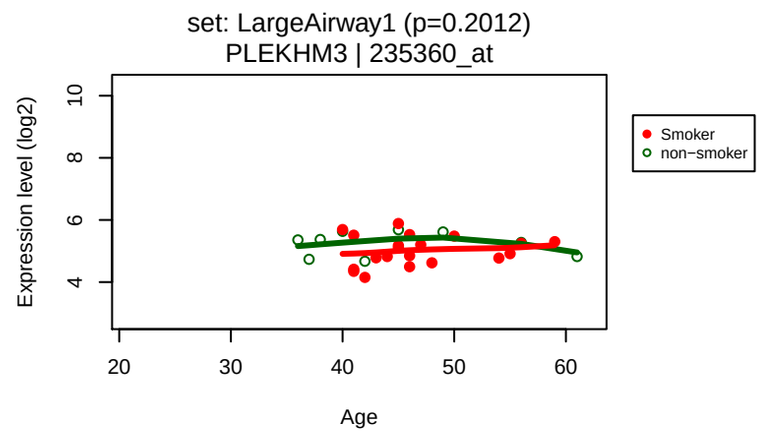

Gene expression of smokers and nonsmokers
gene: PLEKHM3 | 235360_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0157
p=0.1416
p=0.0293
p=0.2012
p=0.7584
p=0.0173
p=0.0431
p=0.0069
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

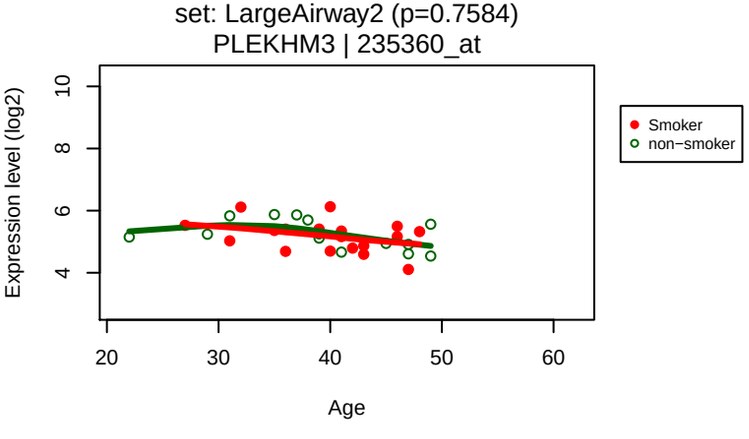

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHM3 | 235360_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0157
p=0.1416
p=0.0293
p=0.2012
p=0.7584
p=0.0173
p=0.0431
p=0.0069
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

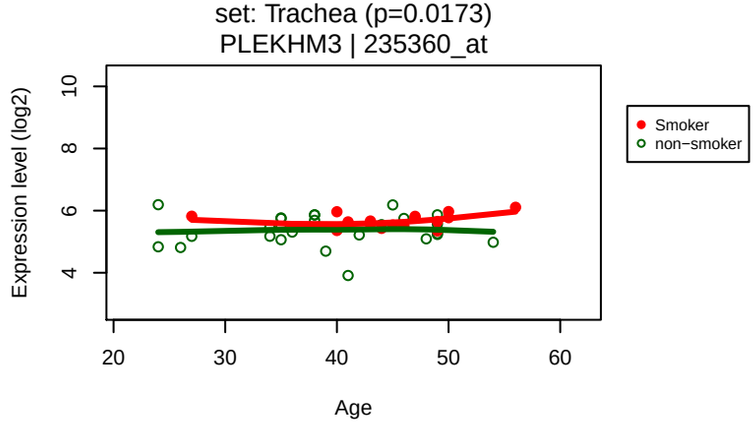

Gene expression ratio in Males
gene: PLEKHM3 | 235360_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6125
p=0.1543
p=0.0607
p=0.006
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHM3 | 235360_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1671
p=0.1215
p=0.2097
p=0.5858
SmallAirway2
LargeAirway2
Trachea
Macrophage2

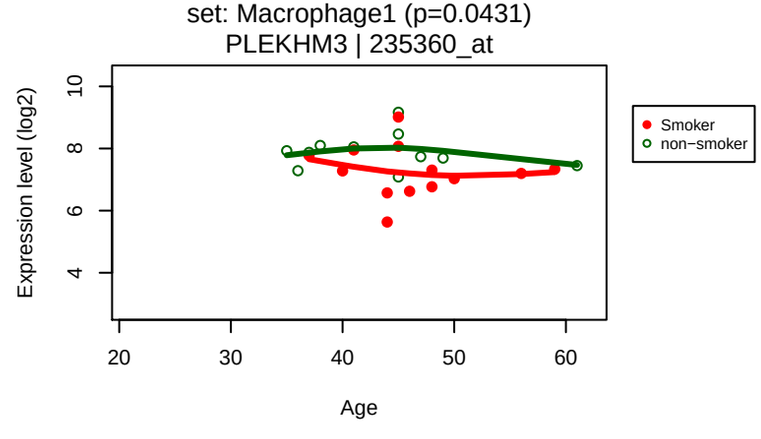

set: SmallAirway1 (p=0.0157)
PLEKHM3 | 235360_at
Expression level (log2)
Age
Smoker
non−smoker

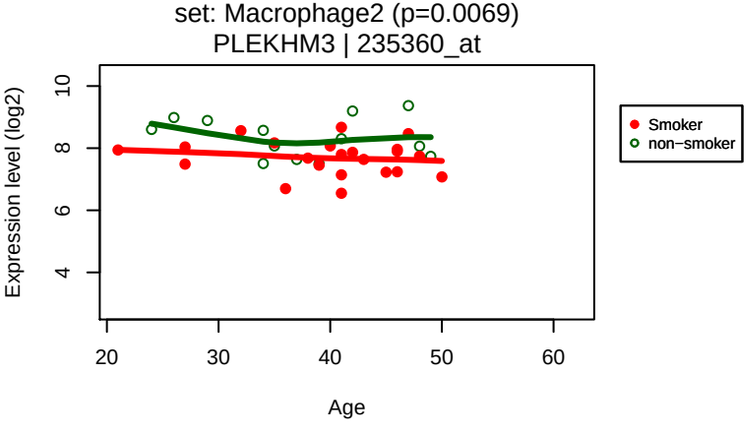

set: SmallAirway2 (p=0.1416)
PLEKHM3 | 235360_at
Expression level (log2)
Age
Smoker
non−smoker

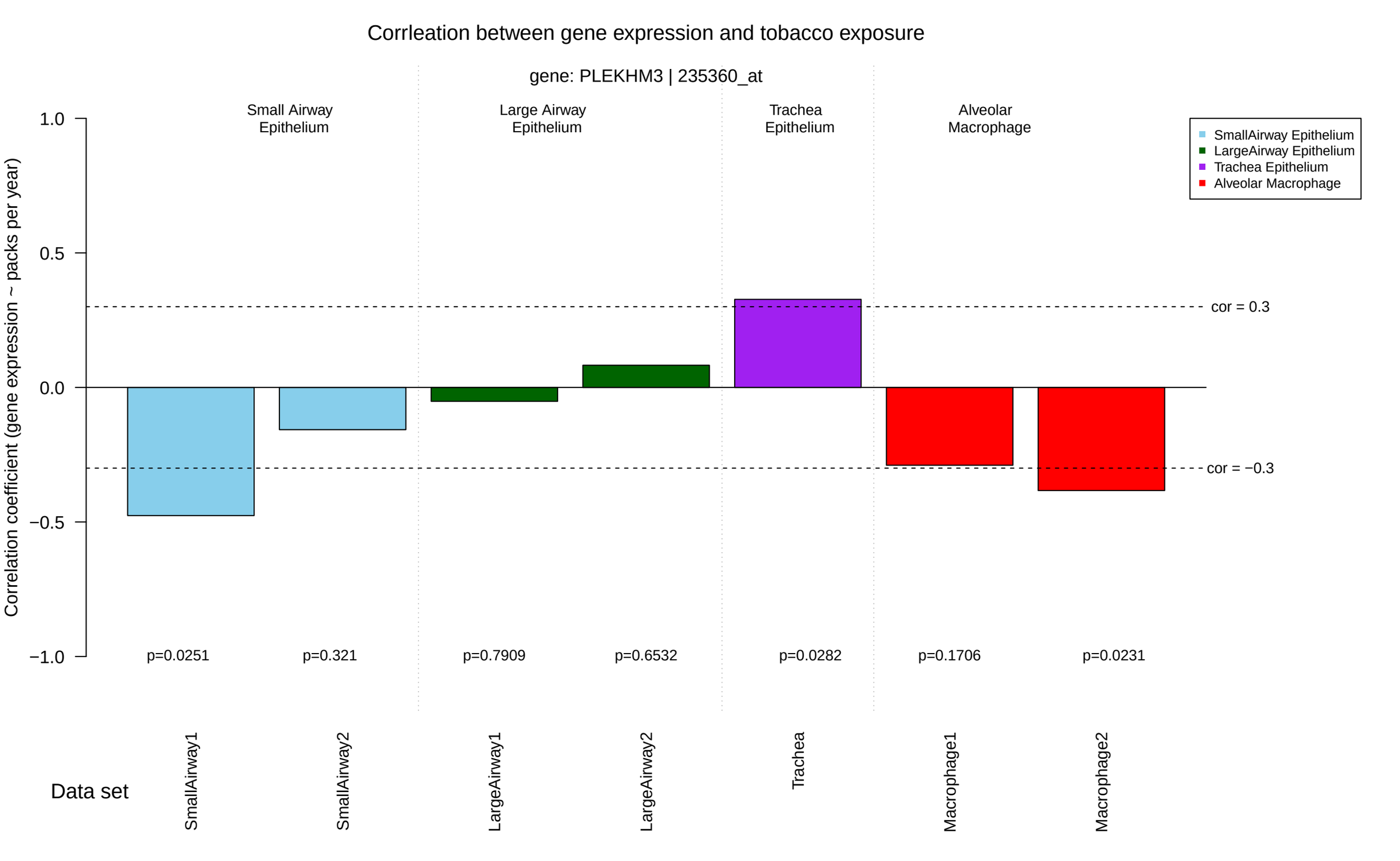

set: LargeAirway1 (p=0.2012)
PLEKHM3 | 235360_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7584)
PLEKHM3 | 235360_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0173)
PLEKHM3 | 235360_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0431)
PLEKHM3 | 235360_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0069)
PLEKHM3 | 235360_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHM3 | 235360_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0251
p=0.321
p=0.7909
p=0.6532
p=0.0282
p=0.1706
p=0.0231
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2